Person A's sections: System, RQ3, Ethics and Limitations (draft for the research rewrite)

IFAS team

I. System

IFAS turns a video into one record per person (Fig. 1). YOLOv8n [1] finds the people in every second frame. A greedy tracker links boxes across frames by intersection over union (IoU) and keeps a lost track open for up to 8 s, so a person the detector misses for a moment is not counted twice. The tracker has no appearance model.

Two classifiers label the crops of each tracked person. Gender comes from a ViT-B/16 [2] with a linear head, fine-tuned on PETA [3]. Age comes from colour and LBP histograms [4], [5] scored by one SVM per age group [6], whose margins a softmax turns into a distribution over the four PETA groups. RQ1 and RQ3 use the released models unchanged, and RQ2 retrains the gender model.

A person is classified at most once every 0.4 s and at most 12 times. Frame i of a person gives a probability vector p_i over the classes $\mathcal{C}$ and a vector of quality cues q_i (Section II). A pooling rule maps the frames $\{(p_i, q_i)\}_{i=1}^{n}$ to one label. The product uses the majority vote

$$\hat{y} = \arg\max_{c \in \mathcal{C}} \sum_{i=1}^{n} \mathbf{1}\Big[\arg\max_{c'} p_i(c') = c\Big], \qquad (1)$$

with a tie going to the class with the larger $\sum_i p_i(c)$. Section II compares it with four other rules and a learned baseline.

II. RQ3: Which Frames Should Count?

A person in a store stays in view for many frames, and the frames differ. Some crops are sharp and whole. Others are blurred by motion, cut off at the edge of the frame or half hidden behind another shopper. The majority vote of Eq. (1) gives every crop the same say. RQ3 asks whether a rule that trusts the good crops more names more people correctly, and whether hand measured image quality does as well as a weighting learned from video labels. Our hypothesis (H3) is that quality weighting beats the majority vote, with the largest gain on people who are truncated or occluded.

A. Pooling Rules

Apart from the majority vote, every rule takes a weighted mean of the frames' probability vectors and picks its largest entry:

$$\bar{p}(c) = \frac{\sum_i w_i \, p_i(c)}{\sum_i w_i}, \qquad \hat{y} = \arg\max_c \bar{p}(c). \qquad (2)$$

The rules differ only in the weights w_i.

- *Single frame.* One frame chosen at random. This is what a system without pooling would report.
- *Majority vote.* Eq. (1), the product's rule.
- *Mean probability.* $w_i = 1$.
- *Confidence weighted.* $w_i = \max_c p_i(c)$, the classifier's own confidence in the frame.
- *Quality weighted.* $w_i = \Pr(\text{correct}_i \mid q_i)$, the chance that frame i is classified correctly given its cues, from the quality model below.

B. Quality Cues and the Quality Model

The cue vector q_i has seven entries, all measured when the crop is exported.

- *Sharpness.* The share of the crop's spectral energy at or above 0.25 cycles per pixel, computed on a 128×128 grey version. Blur and low resolution both remove this band. The variance of the Laplacian [7] is kept as a second sharpness cue.
- *Motion.* Dense Farnebäck optical flow [8] between the crop's frame and the previous one: the mean flow magnitude inside the box minus the median magnitude outside every box, in pixels per frame. Subtracting the background removes camera shake, so what remains is the person's own movement, which is what blurs the crop.
- *Truncation.* 1 if the box touches the frame border.
- *Occlusion.* The box's largest IoU with another person's box in the same frame.
- *Size and detector confidence.* The box's share of the frame area and the detector's score.

The quality model is a logistic regression with an L2 penalty from these cues to whether the classifier got the crop right. The Laplacian variance and the size enter as logarithms and the motion as $\log(1+m)$, because each spans several orders of magnitude. Features are standardised, a cue that could not be measured takes the training mean, and the model is fitted by Newton's method. It is small and readable, and its coefficients show which cue matters (Section II-E).

C. Learned Attention Baseline

Video attribute recognition usually learns how to weigh frames from video labels [9]. A reviewer will rightly ask how measured quality compares with that, so we train the same kind of head on top of the frozen per-frame features of our two classifiers: the 768-dimensional ViT feature that the gender

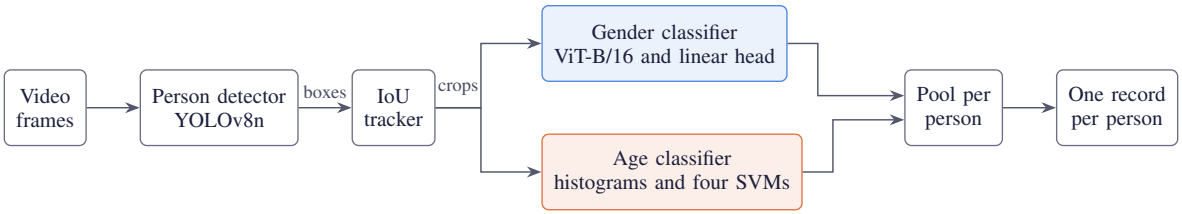

Fig. 1. The IFAS pipeline. Each tracked person is classified on up to 12 frames by both models, and a pooling rule turns those frames into one label per person. RQ3 asks which pooling rule to use.

head reads, and the histograms that the age SVMs score. With f_i the standardised feature of frame i,

$$z_i = W f_i + b, \qquad \alpha_i = \frac{\exp\left(v^\top \tanh z_i\right)}{\sum_j \exp\left(v^\top \tanh z_j\right)}, \tag{3}$$

$$\hat{y} = \arg\max_c \left[U \sum\nolimits_i \alpha_i z_i + u \right]_c, \tag{4}$$

with $z_i \in \mathbb{R}^{32}$. The head is trained with class-balanced cross-entropy and Adam for 300 epochs with a weight decay of 10^{-3}. It needs a label for every training person. Quality weighting needs only crops labelled right or wrong, and that difference in what each method asks for is part of the comparison. MARS [10] with its tracklet attributes [9] is the natural place to train the head. When MARS is not available we train it with the same cross-validation folds as the quality model.

D. Protocol

Two annotators labelled every person in the retail test clips from the crops alone, without seeing the model's answers, and resolved their disagreements together. Agreement before resolution is reported as Cohen's κ [11]. People either annotator marked unsure are left out, and their number is reported.

The per-crop probabilities are the product's own, stored when the crops were exported, so every rule pools exactly what the deployed system saw. The quality model and the attention head are fitted with 5-fold cross-validation grouped by person: no person's crops ever help weigh that person's own frames. The unit of evaluation is the person. We report per-person accuracy with a 95% bootstrap interval over people [12] (2,000 resamples), and McNemar's test [13] of each rule against the majority vote on the same people, exact when fewer than 25 people disagree. To see how much pooling is worth, we also draw $k = 1, \ldots, 12$ of a person's crops at random (20 draws per person) and pool only those. Last, we split people into those who are truncated, or occluded, in at least half their crops, and the rest.

E. Results

Table I and Fig. 2 give the results. With no pooling, a single frame gave a per-person gender accuracy of **??**. The majority vote reached **??** and quality weighting **??** (McNemar $p =$ **??**, with **??** people right only under quality weighting and **??** right only under the vote). For age the same three numbers were **??**, **??** and **??**.

TABLE I
PER-PERSON ACCURACY OF EACH POOLING RULE WITH ITS 95% INTERVAL OVER PEOPLE, AND MCNEMAR'S p AGAINST THE MAJORITY VOTE. GENDER: **??** PEOPLE, AGE: **??** PEOPLE.

Rule	Gender	p	Age	p
Single frame	**??**	**??**	**??**	**??**
Majority vote	**??**	–	**??**	–
Mean probability	**??**	**??**	**??**	**??**
Confidence weighted	**??**	**??**	**??**	**??**
Quality weighted	**??**	**??**	**??**	**??**
Learned attention	**??**	**??**	**??**	**??**

On the **??** people who were truncated in most of their crops, the vote reached **??** and quality weighting **??**. On the **??** occluded people the two reached **??** and **??**.

The learned attention head reached **??** on gender and **??** on age, trained with **??**. Its frame weights and the quality weights had a mean Spearman correlation of **??** for gender.

In the gender quality model fitted on all people, the standardised coefficients were **??** for sharpness, **??** for motion, **??** for truncation and **??** for occlusion. A positive coefficient means crops with more of that cue are classified correctly more often.

Fig. 2b shows how accuracy grows with the number of frames pooled.

III. Ethics

The retail clips are CCTV recordings of members of the public, and the study estimates attributes of those people. We set the following rules before recording anything, and kept to them.

Permission. Every clip comes from a store that gave written permission for this study. The letter says what footage is used, what is done with it, how long it is kept and that the store can withdraw at any time. The footage comes from the store's existing cameras, which are signposted. We installed no camera and recorded nothing new.

Privacy. Footage, crops and labels stay on the project's own machines and are never published or shared. No figure in this paper shows a recognisable person. We do not identify anyone, and the system stores no face template or any other means to find a person again in other footage. Results are reported as totals over people.

Labels. The annotators recorded apparent gender and an apparent age group from the crops. These are judgements

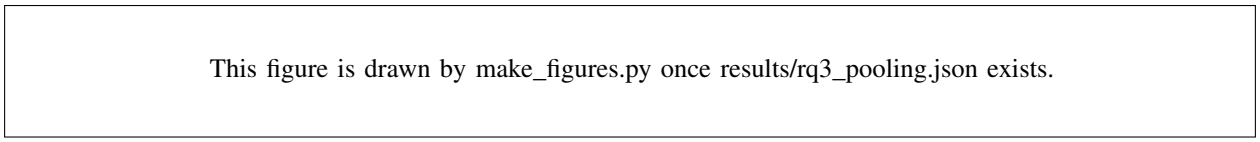

Fig. 2. RQ3 on the retail test clips. (a) Per-person accuracy of each pooling rule with its 95% interval over people. (b) Accuracy when only k of a person's crops are pooled.

about appearance and say nothing about how a person identifies. Gender is recorded as two classes because PETA and MARS label it that way. This is a limitation of the data, and a deployment should report it as an estimate of appearance. Children are not labelled: the age model's youngest group starts at 16, PETA's under-15 group is left out of its training, and no person who looked under 16 in the retail clips was labelled.

Use. The system counts and describes groups of shoppers so a store can plan staffing and layout. It is not meant to treat individual customers differently, and should not be used that way. Gender and age estimates are wrong for a measurable share of people, which Section II reports. They are not fit for any decision about an individual.

IV. Limitations

- *Few stores.* The retail clips come from a small number of cameras in a small number of stores. The intervals in Section II cover the variation between people, not between stores, and a store with different lighting or camera placement may behave differently.
- *Labels are judgements.* The ground truth is what two annotators saw in the crops. People they marked unsure are left out, which removes the hardest cases from every accuracy we report.
- *Correlated frames.* Crops of one person taken 0.4 s apart share lighting, pose and clothing, so their errors are not independent. Pooling cannot fix an error that every frame makes, and accuracy against the number of frames pooled shows how much independent evidence the frames really hold.
- *Hand-chosen cues.* The quality model sees seven cues that we chose. A failure that none of them measures, such as a bag held in front of the body or strong backlight, gets no lower weight.
- *Small training sets.* With MARS unavailable, the quality model and the attention head were fitted on the retail people themselves, with cross-validation. Both would likely improve with more data, and the attention head, with far more parameters, more so.
- *The tracker.* Pooling is only as good as the track it pools. The tracker has no appearance model, so two people who cross paths can swap tracks, and a pooled label can then mix two people.

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